

Atharva Nilawar

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EDUCATION

University of Southern California, Viterbi

Master of Science in Applied Data Science

Aug 2025 - May 2027

GPA: 3.9/4.0

NMIMS University, India

Bachelor of Technology in Data Science

Jun 2021 - Aug 2025

GPA: 3.79/4.0

EXPERIENCE

Data Science Intern

Mercer (Marsh McLennan)

Jan 2025 - Jul 2025

Mumbai, India

- **Quarterly Briefing RAG Platform:** Parsed 100+ quarterly reports (PDF, PPT, Word) into a FAISS vector DB with MongoDB store, auto-generating executive dashboards for new hires, managers, and stakeholders
- On top of this platform, benchmarked RoBERTa, Mistral, and GPT on retrieval accuracy and latency, and shipped a historical-trend chatbot with guardrails, cutting manual review time 90% for 50+ internal users
- **Carrier Survey Report Modernization:** Refactored legacy R codebase powering the firm's flagship carrier survey report into Python for 500+ marketing end-users, cutting production time from 2 weeks to 1 hour
- **Executive Summary Automation:** Automated client-facing Executive Summary PPT generation in Python with version control, cutting marketing report turnaround from 2 weeks to 30 minutes for 50+ end-users

Data Science Intern

Mahindra and Mahindra

May 2024 - Jul 2024

Mumbai, India

- **In-Showroom Humanoid Robot:** Built Azure-hosted chatbot backend using GPT APIs and a reusable RAG pipeline on Mahindra vehicle data, engaging walk-in customers with synchronized vehicle imagery while dealership staff were occupied, improving showroom customer experience
- **Call Center Deflection Bot:** Developed Azure RAG chatbot with intent-routing (dealership locator, service and car-wash booking) to offload volume from live agents, achieving 83% positive feedback across a 500+ participant closed beta
- Validated the same bot's responses via FastAPI and Postman internal testing, enforcing guardrails to block off-topic replies and protect brand voice in customer-facing deployments
- **CMS Portal Automation:** Automated Mahindra CMS portal updates using Selenium and Docker-based workflows, cutting manual refresh cycles from 1 to 2 days down to under 10 minutes for internal content teams

Software Intern

Flowres.IO (myMRPlace)

May 2023 - Jul 2023

Mumbai, India

- **Quant Research Pipeline:** Contributed to a TensorFlow and PyTorch pipeline that ingested analyst Excel inputs, retrieved context via RAG, and called OpenAI APIs to auto-generate structured Q and A outputs
- **Prompt Engineering and QA:** Designed prompt-engineering workflows and ran validation cycles via Postman and Docker to ensure response accuracy before handoff to downstream research consumers

ACADEMIC PROJECTS AND PUBLICATIONS

DRIV: Dynamic Retrieval-based Intelligent Vehicle Recommender

Python, Streamlit, FAISS, Gemini API, GitHub Actions, Selenium

- **Automated Data Pipeline:** Built a monthly Selenium and GitHub Actions workflow scraping Indian automotive sites into a live FAISS vector DB of specs, pricing, and features, removing stale-data risk in the pre-LLM-browsing era
- **Recommendation Engine:** Deployed a Streamlit app with a Gemini-powered chatbot delivering real-time personalized vehicle comparisons, eliminating the need for users to research across multiple dealer sites

Datathon 2.0 Winner: AI Vegetable Detection System

Python, YOLO, OpenCV, Streamlit

Feb 2024

- **Computer Vision System:** Won Datathon 2.0 with a YOLO and OpenCV pipeline identifying vegetables from images in real time, eliminating barcode scanning at retail checkout
- **Retail Deployment Scope:** Shipped a Streamlit demo interface with a roadmap for supermarket rollout covering automated weight estimation and price computation at checkout

Green Technology: Analyzing the Impact of High-Speed Rail in India

Peer-Reviewed IGI Global Book Chapter

2024

- **Peer-Reviewed Publication:** Co-authored an IGI Global chapter evaluating the environmental and economic impact of India's high-speed rail program through data-driven sustainability indicators. <https://doi.org/10.4018/979-8-3693-3985-5.ch003>

TECHNICAL SKILLS

Languages and Databases: Python, SQL, C++, R, JavaScript, MongoDB, FAISS

ML and AI: PyTorch, TensorFlow, Keras, Scikit-Learn, LightGBM, XGBoost, LangChain, Transformers, OpenCV, Pandas, NumPy

Cloud and Tools: Azure, Docker, FastAPI, Selenium, GitHub Actions, Tableau, Power BI, Streamlit, Jupyter, Postman